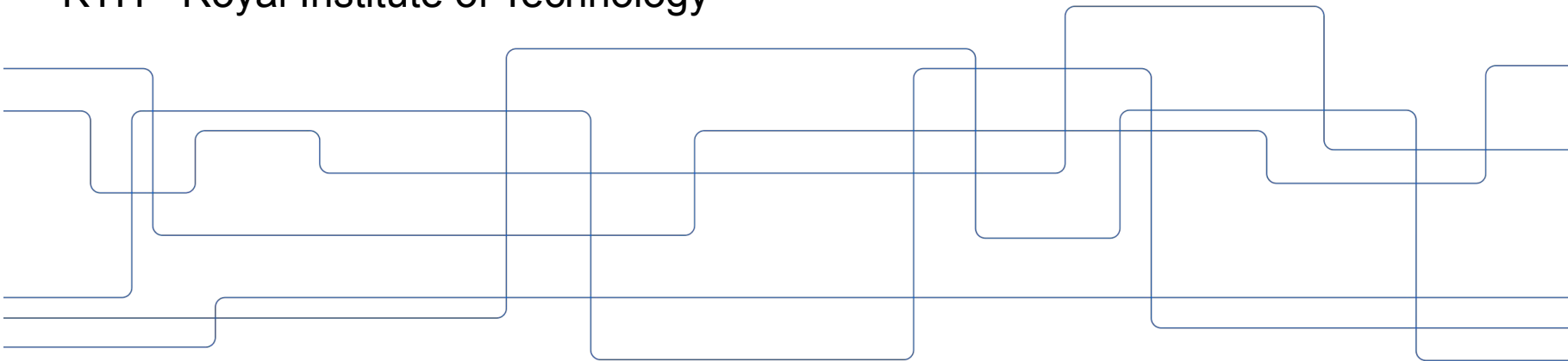




Computer lab- Molecular Simulation

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Presentation – Outline

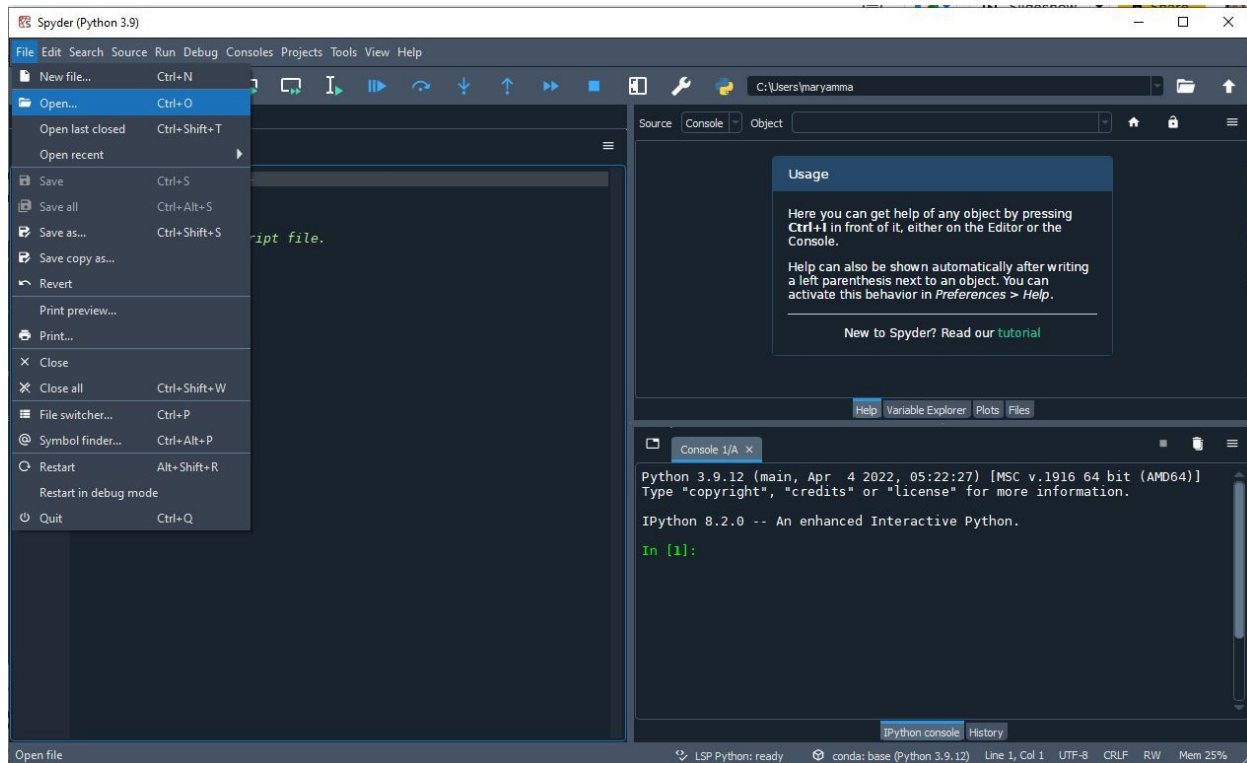
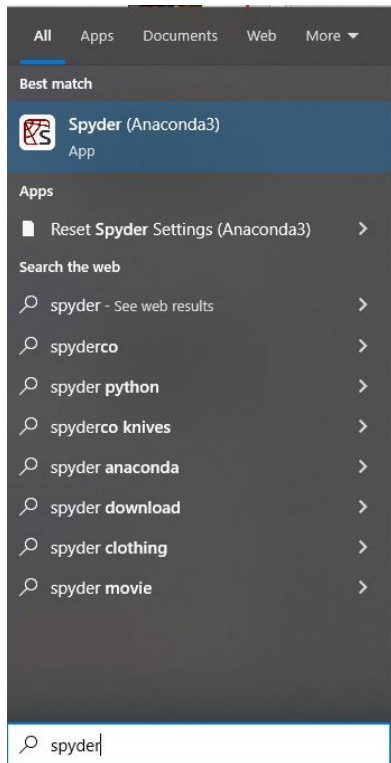
- General description of the Python code and simulation
- Using Spyder to run the simulations
- Report's description
- Report correction and evaluation



General description of the simulation

- Simulate the dynamic behavior of N particles of given mass in 2 dimensions at a given temperature T , for the specified duration integrating the Newton's equation of motion.
 - Download the python script `md_exercise.py` (on CANVAS)
 - Open the script using Spyder

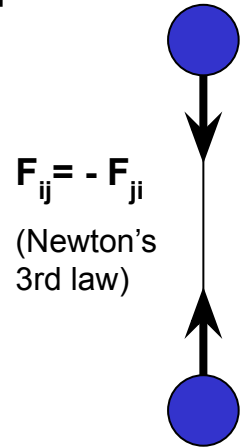
Open the script using Spyder





Description of the simulation and the Python script

- Define $r(0)$ and $v(0)$ for each atom
 - In the initial configuration the particles are at a distance of at least 0.35. The starting velocities are randomly generated according to the given temperature.
- Calculate the configuration of the system over time using Newton's equations of motion
 - $F = ma$
 - $F_{ij} = -F_{ji}$



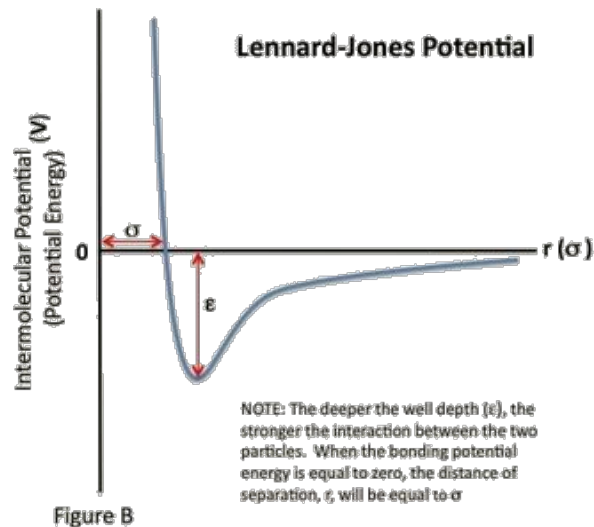
How to calculate force

The particles interact with each other via van der Waals interactions, which we model using the Lennard-Jones potential:

$$V_{VDW}(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right]$$

The parameter ϵ is the well-depth and the parameter σ is the distance at which the potential is zero.

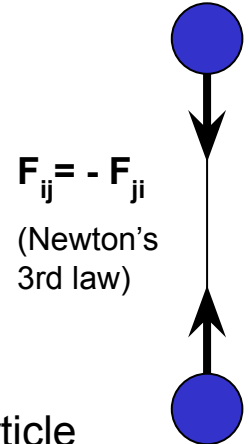
$$F_i = - \frac{\partial}{\partial r_i} V(r_1, r_2 \dots \dots r_N)$$





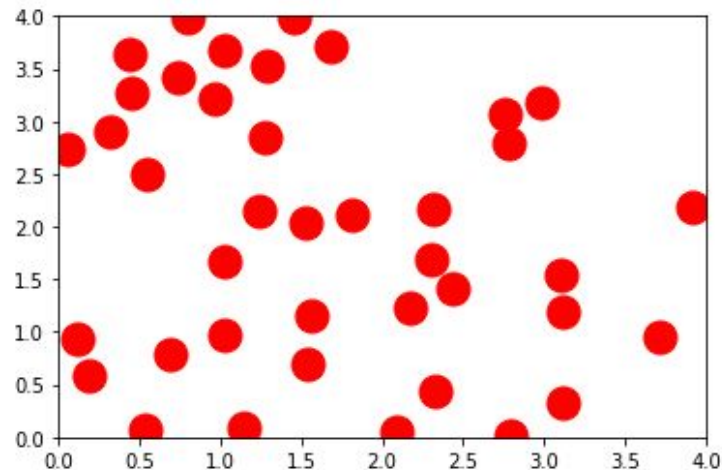
Description of the simulation and the Python script

- Define $r(0)$ and $v(0)$ for each atom
 - In the initial configuration the particles are at a distance of at least 0.35. The starting velocities are randomly generated according to the given temperature.
- Iterate over k steps:
 - Calculate the non-bonded forces based on $r(i)$:
 - Initialize force arrays for each atom
 - Calculate forces applied on atom _{j} due to interaction with atom _{k}
 - Add the force to the array element of atom _{j}
 - Subtract the force from the array element of atom _{k}
 - Integrate to generate new velocities $v(i+1)$ and positions $r(i)$ for each particle
- Total energy of the system ($E_{\text{tot}} = E_{\text{pot}} + E_{\text{kin}}$) should be conserved. Always monitor this during the simulation.



First task description

Simulating the dynamic behavior of **40 Argon atoms** in **2 dimensions** for **50 ps** at a temperature of **300 K** in a box with a length of **4 nm** in a periodic boundary condition and a time step of **0.01 ps**.





Computer lab – Running simulations

- Complete the missing parts in the python script (#TO FILL) following the instruction. The instruction PDF can be found in Canvas
- Run the script using Spyder
- Address the points in the instruction using the script.
- Answer the questions by presenting your results in a report form. Recall to add your name/names to the report and upload it to canvas as soon as possible and before 18th of February 2026.



Computer lab – Complete the python script

Fill the needed parameters
based on the instruction



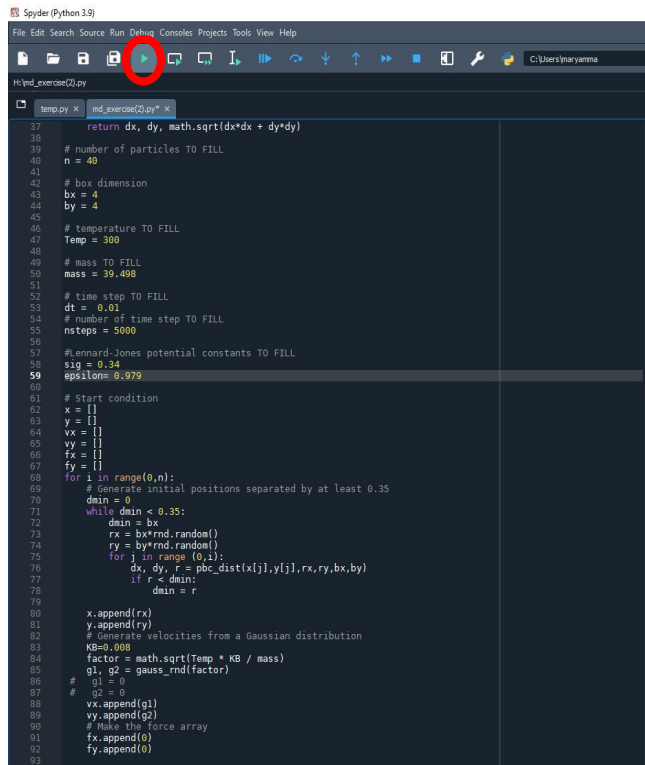
```
H:\md_exercise(2).py
temp.py x md_exercise(2).py x

16
17 # The potential
18 def V(r):
19     return 4*epsilon*((math.pow(sig,12)*math.pow(r,-12)) - (math.pow(sig,6)*math.pow(r,-6)))
20
21 # The force
22 def F(r):
23     return 4*epsilon*((12*math.pow(sig,12)*math.pow(r,-13)) - (6*math.pow(sig,6)*math.pow(r,-7)))
24
25 # Calculate the shortest periodic distance, unit cell [0,bx],[0,by]
26 def pbc_dist(x1,y1,x2,y2,bx,by):
27     dx = x1 - x2
28     dy = y1 - y2
29     if dx < -0.5*bx:
30         dx += bx
31     if dx > 0.5*bx:
32         dx -= bx
33     if dy < -0.5*by:
34         dy += by
35     if dy > 0.5*by:
36         dy -= by
37     return dx, dy, math.sqrt(dx*dx + dy*dy)
38
39 # number of particles TO FILL
40 n =
41
42 # box dimension
43 bx = 4
44 by = 4
45
46 # temperature TO FILL
47 Temp =
48
49 # mass TO FILL
50 mass =
51
52 # time step TO FILL
53 dt =
54 # number of time step TO FILL
55 nsteps =
56
57 #Lennard-Jones potential constants TO FILL
58 sig =
59 epsilon=
60
61 # Start condition
62 x = []
63 y = []
64 vx = []
65 vy = []
66 fx = []
67 fy = []
68 for i in range(0,n):
69     # Generate initial positions separated by at least 0.35
70     dmin = 0
71     while dmin < 0.35:
72         dmin = bx
```



Computer lab – Run the script using Spyder

Run the script



```
37     return dx, dy, math.sqrt(dx*dx + dy*dy)
38
39 # number of particles TO FILL
40 n = 40
41
42 # box dimension
43 bx = 4
44 by = 4
45
46 # temperature TO FILL
47 Temp = 300
48
49 # mass TO FILL
50 mass = 39.498
51
52 # time step TO FILL
53 dt = 0.01
54 # number of time step TO FILL
55 nsteps = 5000
56
57 # Lennard-Jones potential constants TO FILL
58 sig = 0.34
59 epsilon = 0.979
60
61 # Start condition
62 x = []
63 y = []
64 vx = []
65 vy = []
66 fx = []
67 fy = []
68 for i in range(0,n):
69     # Generate initial positions separated by at least 0.35
70     dmin = 0
71     while dmin < 0.35:
72         dmin = bx
73         rx = byrnd.random()
74         ry = byrnd.random()
75         for j in range(0,i):
76             dx, dy, f = pbc_dist(x[j],y[j],rx,ry,bx,by)
77             if f < dmin:
78                 dmin = f
79
80     x.append(rx)
81     y.append(ry)
82
83     # Generate velocities from a Gaussian distribution
84     KB=0.008
85     factor = math.sqrt(Temp * KB / mass)
86     g1, g2 = gauss_rnd(factor)
87     # g1 = 0
88     # g2 = 0
89     vx.append(g1)
90     vy.append(g2)
91     # Make the force array
92     fx.append(0)
93     fy.append(0)
```





Point to address to the report

a) The atoms interact with van der Waals interactions, which we model using the Lennard-Jones potential. As was explained before, the parameter ϵ is the well-depth. Perform a simulation using a Lennard-Jones potential with a depth of 1.5 kJ/mol (keep the initial σ value) and compare the results with the simulations you have already performed for Argon atoms. In particular:

- Draw the potential energy curve in function of the distance and compare the shapes. Add a plot with both curves in it and explain what has changed.
- Describe the changes in the energies derived from the Python simulations. Support your analysis with suitable graphs.



Point to address to the report

b) Total energy ($E_{\text{tot}} = E_{\text{pot}} + E_{\text{kin}}$) in the simulation should be conserved, so it is good practice to monitor this during the simulation.

- Describe what happens to the total energy if you use a time step larger than 0.01 ps.
- What is the largest time step that one can use to simulate the Argon atoms and keep energy conservation. Support your results with simulation outputs.



Point to address to the report

c) Describe how temperature and mass affect the kinetic energy.

- To achieve this, perform simulations using different values for temperature and mass.
- Tips: change only one value in each simulation.



Report's description

- Answer the provided questions and upload the answers as a report (Max 1500 words) before 18th of February 2026 in Canvas.
- Address each point in a separate session.
- Provide all the necessary explanation in a clear way.
- Plots/Figure should have caption, legend and labels on the axis. Values should be written together with unit.
- Only one resubmission is possible. Only if the issue is addressed properly in the first resubmission will you pass the lab.



Report corrections:

- Criteria of evaluation
 - Report should be clearly organized
 - All the points are correctly addressed (no wrong statement)
- Procedure
 - Deadline 18th of February
 - Feedback via Canvas
 - All the discussion concerning the report should take place in canvas (not via email)
 - Resubmission should be done within 1 week from feedback

Questions?

